Research question "What design approaches and technological strategies are currently used to enable personalization and user involvement in upper limb prosthetic design, and what limitations and research gaps remain?" Review protocol "Conduct a systematic literature review of peer-reviewed academic research on upper limb prostheses and assistive devices, focusing on design-oriented perspectives rather than purely clinical outcomes. Include studies published from 2005 onwards that address at least one of the following dimensions: – industrial design, inclusive design, or user-centred design approaches; – additive manufacturing and parametric design for customization; – use of anthropometric data in product adaptation; – artificial intelligence applied to design processes (not diagnosis); – digital platforms, configurators, or user-configurable systems for prosthetics or assistive products. Exclude studies focused exclusively on surgical procedures, rehabilitation therapy outcomes, biomechanical signal processing, or AI for clinical diagnosis without relevance to design workflows." Data extraction focus "For each included study, extract: – the design paradigm or framework used; – the role of the user in the design or configuration process; – the technological approach adopted; – reported limitations or challenges." Synthesis objective "Synthesize the literature to identify: – converging trends across design, fabrication, and digital systems; – recurring limitations in current approaches; – clear research gaps that justify further investigation into AI-assisted, parametric, and user-configurable prosthetic design platforms."

Current upper limb prosthetic personalization relies predominantly on user-centered and participatory design frameworks coupled with 3D printing, CAD systems, and 3D scanning technologies, but persistent limitations in material durability, lack of standardized testing protocols, and high device abandonment rates reveal critical research gaps in AI-assisted design optimization, comprehensive user-configurable functional platforms, and long-term outcome validation.

Abstract

User-centered design represents the dominant framework for upper limb prosthetic personalization, employed in approximately half of the 26 reviewed studies, with participatory design approaches enabling direct user involvement through co-design sessions and feedback mechanisms. 3D printing has achieved near-universal adoption as the primary manufacturing technology, typically integrated with CAD systems and 3D scanning for anatomical data capture. Material costs remain low—under $50 per prosthesis in some implementations—and patient satisfaction ex-

ceeding 90% has been reported in humanitarian contexts employing participatory approaches. User involvement varies substantially, ranging from comprehensive participation across all design stages to evaluation-only engagement, with studies demonstrating that early and sustained user participation reduces abandonment risk.

Persistent limitations constrain current approaches, including material durability failures at high-stress regions, absence of standardized testing protocols for 3D-printed prosthetics, and high abandonment rates linked to comfort, functionality, and aesthetic concerns. Critical research gaps remain in AI integration for design optimization—only one study implemented machine learning for prosthetic interaction—while digital platforms enabling user configuration are limited to aesthetic selection portals rather than comprehensive functional customization systems. Long-term durability and retention data are largely absent, and the integration of emerging technologies including AI, IoT, and wireless communication with user-centered design methodologies represents substantial unexplored territory. These gaps support the need for investigation into AI-assisted, parametric, and user-configurable prosthetic design platforms that systematically incorporate validated user involvement throughout the design lifecycle.

Paper search

We performed a keyword search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: `("upper limb prosthetic" OR "upper limb prosthesis" OR "arm prosthetic" OR "hand prosthetic") AND (design OR customization OR personalization OR "user involvement" OR "user-centered design" OR "user-centred design") AND ("industrial design" OR "inclusive design" OR "universal design" OR "additive manufacturing" OR "3D printing" OR "parametric design" OR anthropometric OR anthropometry OR "artificial intelligence" OR "AI-assisted design" OR "digital platform" OR configurator OR "configurable system") AND year:[2005 TO *]`

The search returned 52 total results from Elicit.

We retrieved 52 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Core Theme Alignment**: Does this study clearly address at least one core theme: design approaches, customization/personalization, or user involvement in prosthetic development?
- **Upper Limb Prosthetics Focus**: Does this study focus on upper limb prosthetics or closely related assistive devices with clear relevance to upper limb design?
- **Design Relevance Over Clinical Focus**: Is the primary focus of the study related to design processes, systems, or development strategies rather than exclusively to surgical or rehabilitation outcomes?
- **Design or Customization Approach**: Does this study address design-oriented approaches (e.g., industrial design, user-centered design) or technological strategies for customization or personalization (e.g., additive manufacturing, parametric design)?
- **User-Centered or Anthropometric Elements**: Does this study involve user participation in the design process or the use of anthropometric data to support prosthetic adaptation?
- **Digital Tools or AI in Design**: Does this study involve digital platforms, configurators, user-configurable systems, or AI applications within prosthetic design workflows (excluding clinical diagnosis or signal processing)?

- **Explicit Discussion of Limitations or Gaps**: Does the study explicitly discuss limitations, challenges, or unresolved issues related to prosthetic design, customization, or user involvement?
- **Publication Quality**: Is this a peer-reviewed academic publication (journal article, systematic review, or meta-analysis)?

Papers that failed any strict criterion were automatically excluded. For the remaining papers, we considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Design Framework**:

  Extract the design paradigm, approach, or methodology used for upper limb prosthetic development, including:

- Specific design framework (user-centered design, inclusive design, participatory design, etc.)
- Design process steps or stages described
- Design principles or guidelines followed
- Whether interdisciplinary approaches were used and which disciplines were involved
- Any design standards or protocols referenced -If elements are not explicitly stated in the paper, indicate 'not specified.

- **User Involvement**:

  Extract details about how users (prosthetic users/amputees) participated in the design or configuration process, including:

- Stage(s) of involvement (needs assessment, design phase, prototyping, testing, evaluation)
- Methods of user participation (interviews, co-design sessions, testing, feedback)
- Level of user control over design decisions
- Number and characteristics of users involved
- User input mechanisms (physical interaction, digital interfaces, surveys)
- How user preferences, needs, or feedback influenced the final design
- Extract explicitly reported forms of user involvement only. Do not infer user participation if not stated.

- **Technology Strategy**:

  Extract the technological approaches and tools used to enable personalization and customization in upper limb prosthetic design, including:

- Manufacturing technologies (3D printing, additive manufacturing, traditional manufacturing)
- Digital design tools (parametric design, CAD systems, configurators)
- AI/machine learning applications for design (not clinical diagnosis)
- Data collection methods (anthropometric measurement, 3D scanning)
- Digital platforms or software interfaces for user configuration
- Integration of multiple technologies and how they work together
- Distinguish between technologies used as core enablers of the design process and those used only as supporting tools.

- **Customization Features**:

  Extract what aspects of the upper limb prosthetic can be personalized or customized, including:

- Physical dimensions and fitting (socket design, sizing, ergonomics)
- Functional features (grip patterns, control mechanisms, gestures)
- Aesthetic elements (appearance, colors, textures, cosmetic covers)
- Modular components that can be adjusted or swapped
- User-configurable settings or parameters
- Range of personalization options available
- Whether customization is one-time or ongoing/adaptive

- **Study Limitations**:

  Extract limitations, challenges, and barriers reported in the study related to personalization and user involvement in upper limb prosthetic design, including:

- Technical limitations of current approaches
- Manufacturing constraints or cost issues
- User adoption or acceptance challenges
- Design process limitations or inefficiencies
- Technology limitations (materials, software, hardware)
- Scalability or accessibility barriers
- Time or resource constraints
- Integration challenges between different approaches or technologies
- if an article does not discuss limitation write not discussed

- **Research Gaps**:

  Extract identified research gaps, future research directions, and recommendations specifically related to personalization and user involvement in upper limb prosthetic design, including:

- Areas needing further investigation
- Suggested improvements to current approaches
- Future technology integration opportunities
- Unmet user needs or design requirements
- Calls for better design methodologies or frameworks
- Recommendations for AI-assisted, parametric, or user-configurable platforms
- Suggestions for addressing current limitations
- Areas where more user involvement is needed

- **Contribution Type**:

  What is the primary contribution of the study (conceptual framework, methodology, system/platform, case study, technical prototype)?

Results

Characteristics of Included Studies

This review synthesized findings from 26 studies published between 2016 and 2025, examining design approaches and technological strategies for personalization and user involvement in upper limb prosthetic design. The studies encompassed a range of contribution types including technical prototypes, case studies, conceptual frameworks, and methodological developments.

Study	Full Text Retrieved?	Contribution Type	Primary Design Framework	Core Technology
Pérez Dickson et al., 2019	No	System/platform, methodology	Design Sprint	3D printing
A. Hussaini et al., 2023	Yes	Case study	Participatory design, user-centered design	3D printing, Fusion 360 CAD
Kai Xu et al., 2023	Yes	Conceptual framework	Interdisciplinary collaboration, participatory design	3D printing, CAD software
Jennifer Mankoff et al., 2018	Yes	Methodology	Participatory design	Additive manufacturing, online survey tool
Pierre Moreau et al., 2020	No	Case study	User-centered design	3D printing, 3D scanning, telemedicine
Julio Fajardo et al., 2017	No	Technical prototype	User-centered design	3D printing, parametric design, sEMG control
Bryce Dyer et al., 2022	Yes	Case study, technical prototype	Dyer framework (2018)	3D printing, CAD (SolidWorks, 3D Studio Max)
Julio Fajardo et al., 2020	No	Technical prototype	User-centered design, parametric design	3D printing, parametric design, EMG electrodes
K. Wendo et al., 2025	No	Methodology	Not specified	3D printing, REDCAP platform
Alejandro M. Saldarriaga et al., 2024	No	Methodology, technical prototype	Parametric design	3D printing with PLA, parametric design
M. Hofmann et al., 2016	Yes	Case study	Participatory design	3D printing, anthropometric measurement
Amanda Coelho Figliolia et al., 2019	No	Case study	User-centered design	3D printing

Study	Full Text Retrieved?	Contribution Type	Primary Design Framework	Core Technology
Albert Manero et al., 2023	Yes	Technical prototype	User-centered, participatory, inclusive design	Additive manufacturing, thermoforming, web portal
A. Roda-Sales et al., 2024	Yes	Case study, conceptual framework	User-centered design	3D printing, 3D scanning, CAD/CAM
Ricardo Rodrigues et al., 2025	Yes	Technical prototype	User-centered design, inclusive design	3D printing, EMG sensors, Arduino
F. Pinheiro et al., 2018	No	Technical prototype	Not specified	Additive manufacturing, CAD, flexible TPU filament
Guisheng Xu et al., 2017	No	Case study	Not specified	3D printing
Wilson Sutanto Tan et al., 2024	No	Technical prototype	Not specified (user-centered implied)	3D printing (FDM), 3D scanning, CAD
Renato Mio et al., 2019	Yes	Case study	Not specified	FDM 3D printing, CAD
Jung Wook Park et al., 2023	Yes	Technical prototype	Not specified	Multi-material 3D printing, Fusion 360
Andrea Demofonti et al., 2022	Yes	Technical prototype	User-centered and inclusive design (implied)	CAD systems, modular design
Muhammad Firas Zulkifli et al., 2023	Yes	Technical prototype	User-centered design	3D printing, 3D scanning
A U Iwuoha et al., 2020	No	Technical prototype	Not specified	3D printing, Fusion 360 parametric design
Juan Sebastian Cuellar et al., 2018	Yes	Case study, technical prototype	Ten design considerations for AM	FDM 3D printing (Ultimaker 3)
Maxwell Salazar et al., 2024	Yes	Technical prototype, methodology	User-centered design	FDM 3D printing, CAD (Fusion360)
Saeed Bahrami Moqadam et al., 2025	No	Conceptual framework	Human-computer interaction framework	AI (BiLSTM algorithm), material detection sensors

The included studies represent diverse geographic contexts, with several specifically addressing low- and middle-income country (LMIC) settings. Technical prototypes constitute the largest category (15 studies), followed by case studies (9 studies), with some studies contributing both prototype development and methodological frameworks.

Full texts were retrieved for 15 of the 26 studies, with the remainder analyzed through detailed abstracts.

Design Frameworks and Approaches

Prevalence of Design Paradigms

User-centered design emerged as the dominant framework, explicitly adopted in 13 studies. Participatory design, which emphasizes direct user involvement in design activities, was employed in five studies. Parametric design approaches, enabling systematic customization through adjustable parameters, featured in four studies.

Design Framework	Studies Employing Framework	Key Characteristics
User-centered design	13 studies	Iterative design optimization with patient feedback, exploration of user needs and preferences, systematic approach focusing on ergonomics and usability
Participatory design	5 studies	Co-design activities, direct user engagement in design sessions, web-based customization portals
Parametric design	4 studies	Modular and self-contained design, optimization of design time and computational load
Interdisciplinary collaboration	3 studies	Integration of engineering, materials science, biomechanics, and healthcare, collaboration with industry consultants
Inclusive design	3 studies	Focus on accessibility and affordability, holistic design decisions
Design Sprint	1 study	Step-by-step validation of design decisions
HCI-based framework	1 study	Human-computer interaction for auditory feedback
Not specified	6 studies	Technology-focused without explicit design framework articulation

Interdisciplinary collaboration was identified as critical for effective prosthetic design, with studies emphasizing integration across engineering, materials science, biomechanics, healthcare, artificial intelligence, and virtual reality. The MSF humanitarian project exemplified this approach through a multidisciplinary team spanning prosthetics, orthotics, physical and occupational therapy, rehabilitation medicine, surgery, and biomedical engineering.

Design Principles and Guidelines

Several studies articulated specific design principles guiding their development processes. Simplicity without neglecting performance in terms of grasping capabilities, power consumption, and controllability was emphasized.

Design considerations for additive manufacturing of non-assembly mechanisms were formalized into ten guidelines addressing body-powered control, adaptive grasp, cosmetics, low weight, and waterproofing. For sports prosthetics, principles included reduction in aerodynamic drag, appropriate aspect ratios, and forms allowing for movement and asymmetry. The wrist module development prioritized lightness, compactness, dexterity, ease of use, reliability, and robustness.

Technological Strategies for Personalization

Manufacturing Technologies

3D printing emerged as the predominant manufacturing technology, utilized in 24 of 26 studies. Fused Deposition Modeling (FDM) represented the most common 3D printing technique, with multi-material printing capabilities expanding design possibilities.

Technology Category	Specific Technologies	Implementation Examples
Additive manufacturing	FDM 3D printing, multi-material printing, open-source desktop printers	PLA material for sockets, flexible TPU filament, impact-resistant materials
Digital design tools	CAD systems (Fusion 360, SolidWorks, 3D Studio Max)	Parametric design for customization, conversion to STL format
Data acquisition	3D scanning, anthropometric measurement	Scanning accuracy of ±2 mm, laser scanning methods
Control interfaces	sEMG/EMG sensors, graphical user interfaces	Arduino-based signal processing, threshold-based activation
Digital platforms	Web-based customization portals, telemedicine, REDCAP, KoboToolbox	Remote digital design support, user feedback collection
AI/ML applications	BiLSTM algorithm, CNN (planned)	Material detection classification at 92.46-97.57% accuracy, future multi-DoF control

Material costs for 3D-printed prostheses were reported as less than $50 per upper-limb prosthesis including trial components, less than $20 for transparent facial orthoses, and less than $20 for pediatric wrist prostheses. Manufacturing times were notably rapid, with one case reporting less than 8 hours for printing and 20 minutes for assembly.

Integration of Multiple Technologies

Studies increasingly combined multiple technologies to enhance personalization capabilities. The integration of 3D scanning with CAD/CAM systems enabled accurate anatomical fitting. Telemedicine facilitated remote digital design and expert clinical support in humanitarian contexts. The combination of additive manufacturing with thermoforming achieved a 33% reduction in cosmesis component weight. Real-time monitoring systems integrated hardware and software components for performance optimization.

User Involvement in Design Process

Stages and Methods of User Participation

User involvement varied substantially across studies, ranging from comprehensive co-design activities to evaluation-only participation.

Study	Stages of Involvement	Methods Used	Users Involved
A. Hussaini et al., 2023	Needs assessment, design, prototyping, testing	Co-design sessions, interviews, KoboToolbox	21 participants (Co-design 1), 17 participants (Co-design 2)
Pierre Moreau et al., 2020	Needs assessment, design, testing, evaluation	OPUS surveys, remote communication	29 upper limb patients, 24 facial orthosis patients
M. Hofmann et al., 2016	Needs assessment, design, prototyping, testing	Interviews, co-design sessions, testing, feedback	3 participants with congenital amputations
Bryce Dyer et al., 2022	Needs assessment, design, prototyping, testing	Feedback on concepts, testing developmental models	Single male paracyclist
Albert Manero et al., 2023	Visual design phase	Web-based customization portal, color/design selection	Pediatric patients and parents/guardians
A. Roda-Sales et al., 2024	Needs assessment	Anonymous online questionnaire	18 participants with upper limb conditions
Muhammad Firas Zulkifli et al., 2023	All stages (needs assessment through evaluation)	3D scanning, measurements, feedback sessions	Single patient
Saeed Bahrami Moqadam et al., 2025	Evaluation	Not specified	10 participants (3 amputees, 2 blind amputees, 5 controls)

Studies employing participatory approaches reported that user feedback directly drove design decisions, with participants having significant control in defining parameters and making adjustments. The Uganda-based co-design activities identified key design factors including aesthetics and functionality preferences, with users expressing preference for darker brown colors for cosmetic acceptance.

Impact of User Involvement on Design Outcomes

User participation demonstrably influenced final designs. In the case study approach, user involvement led to customized products matching user preferences and minimizing product abandonment risk. Web-based customization portals enabling aesthetic selection increased emotional connection and reduced rejection rates. The survey-based study revealed that users prioritize breathability, low cost, stable fixing systems, lightweight construction, and comfort.

Customization Features

Physical and Functional Customization

The reviewed studies addressed multiple dimensions of prosthetic customization, with physical fitting and functional features receiving the most attention.

Customization Dimension	Examples from Studies	Implementation Approach
Physical fitting	Socket design based on 3D scanning, parametric sizing, ±2 mm accuracy to intact limb	Scanning of residual limb and torso, anthropometric measurements
Functional features	Customizable grip patterns and gestures, energy return elements, interchangeable terminal devices	sEMG control, graphical interfaces for gesture selection
Aesthetic elements	Color selection through web portals, preference for non-human skin tones, surface finish	CAD design for appearance, thermoforming for cosmesis
Modular components	Swappable energy return elements, detachable shoulder/arm via slot-in mechanism, magnetic attachments	Modular design for compatibility across prostheses
User-configurable settings	Real-time monitoring for optimization, adjustable harness systems, graphical interface for gestures	Digital interfaces enabling ongoing adjustment

The modular, parametric, and self-contained design approach enabled fitting across a wide range of transradial amputation levels. One study reported that energy return elements could be exchanged or replaced to adjust energy storage and force generation, with real-time monitoring allowing users to optimize performance.

Customization Temporality

Customization approaches ranged from one-time fitting to ongoing adaptive systems. Iterative design processes with patient feedback characterized ongoing customization, while some studies focused on initial parameter-based customization without subsequent adaptation.

Limitations and Challenges

The reviewed studies identified numerous limitations across technical, manufacturing, user adoption, and design process domains.

Technical and Manufacturing Limitations

Limitation Category	Specific Challenges	Studies Reporting
Print failures and material issues	Print failures with flexible materials, delamination at high-stress regions, mechanical wear of components	Pierre Moreau et al., 2020
Material durability	Properties not as robust as traditional materials, susceptibility to cyclical loading failure, limited durability of FDM molds (50 pulls)	Albert Manero et al., 2023, Maxwell Salazar et al., 2024
Manufacturing constraints	Long printing times, need for stable power supply, time-consuming support removal, significant post-processing time	A. Hussaini et al., 2023, Bryce Dyer et al., 2022
Control limitations	Noise sensitivity and electrode dependency, inability to control fingers independently, threshold-based activation limitations	Ricardo Rodrigues et al., 2025, A U Iwuoha et al., 2020
Standardization gaps	Lack of standards for fabrication and testing, existing standards not applicable to multi-fingered prostheses	Renato Mio et al., 2019

Resource requirements represented a significant barrier. While raw material costs remained low, abundant human and technical resources were required for implementation. The need for optimization of 3D printing parameters demanded countless hours from technical teams.

User Adoption and Design Process Challenges

High abandonment rates due to comfort, cost, and functionality issues were reported, with one study noting over 70% dissatisfaction and 52% abandonment rates among upper limb prosthetic users due to limited functionality, poor design/aesthetic, and improper fit. Complex operation and interaction methods influenced users to disfavor devices. Socket components frequently caused discomfort in both traditional and 3D-printed prosthetics, contributing to high rejection rates.

Design process limitations included geographical dispersion and logistical challenges in user involvement, focus on specific pain points rather than broad activities of daily living, and limited generalizability from single-patient studies. The lack of long-term follow-up data constrained understanding of device durability and user satisfaction over time.

Scalability and Accessibility Barriers

Infrastructure and resource constraints for mass customization were identified, alongside challenges in navigating regulatory and reimbursement policies. Limited access to stable power supply in LMICs impeded 3D printing implementation. Difficulty in sourcing materials locally and ensuring sustainable reproduction and maintenance presented ongoing barriers. Recruiting sufficient sample sizes for research validation remained a perennial challenge.

Synthesis

Explaining Heterogeneity in Design Approaches

The variation in design frameworks across studies reflects differences in project context, resource availability, and target populations rather than fundamental disagreement about best practices. Studies conducted in LMIC humanitarian contexts emphasized participatory co-design with local researchers and users to ensure cultural acceptability and sustainable maintenance, while studies in high-resource settings more frequently adopted technology-driven prototyping approaches without explicit user involvement frameworks.

The apparent tension between comprehensive user-centered approaches and rapid technical prototyping can be reconciled through examination of project goals. Studies targeting immediate humanitarian needs prioritized functional adequacy and local manufacturability, achieving patient satisfaction exceeding 90% with wear times over 5 hours daily. Studies aimed at advancing technical capabilities focused on expanding functional possibilities—such as multi-gesture control or energy return mechanisms—with user validation planned for subsequent phases.

Converging Trends

Despite methodological diversity, several convergent trends emerged across the literature. First, 3D printing has become nearly ubiquitous (24 of 26 studies), establishing additive manufacturing as the dominant fabrication approach for personalized prosthetics. Second, integration of 3D scanning with CAD systems for anatomical data capture and parametric design is becoming standard practice. Third, there is growing recognition that user involvement throughout the design process—not merely at evaluation—reduces abandonment risk and improves satisfaction.

Persistent Limitations Requiring Resolution

Material durability remains problematic across studies, with mechanical failures at high-stress regions, delamination under cyclical loading, and insufficient robustness compared to traditional materials. The absence of standardized testing procedures for 3D-printed prostheses prevents systematic comparison and quality assurance. User involvement, while increasingly recognized as essential, frequently occurs only during evaluation phases rather than throughout design development.

Context-Specific Applicability

The evidence suggests that optimal design approaches depend on implementation context:

- **LMIC humanitarian settings**: Participatory co-design with local stakeholders, simple passive designs with interchangeable components, telemedicine support for technical expertise, and low-cost desktop 3D printers yield functional outcomes with mean patient satisfaction exceeding 90%.
- **Specialized applications (sport, pediatric)**: User-specific scanning and parametric CAD modeling, iterative testing in real-world conditions, and modular aesthetic customization address performance and acceptance requirements.
- **Technology development contexts**: Multi-material printing, AI-assisted control systems, and integrated sensing expand functional capabilities, though real-world validation with end users remains largely pending.

Research Gaps Justifying Further Investigation

The synthesis identifies several critical gaps warranting investigation. First, AI integration in prosthetic design processes remains nascent, with only one study implementing machine learning for material detection and others noting AI as a future direction. No studies demonstrated AI-assisted parametric design optimization or user-configurable platforms leveraging machine learning. Second, digital platforms enabling end-user configuration are limited to a single web-based customization portal for aesthetic selection; no studies reported comprehensive user-configurable systems addressing functional parameters. Third, long-term durability and user retention data are lacking, with most studies limited to short-term evaluation periods. Fourth, standardized testing protocols for 3D-printed prosthetics require development to enable systematic comparison. Fifth, the integration of user-centered design with emerging technologies (AI, IoT, wireless communication) presents substantial unexplored territory.

These gaps collectively support the need for research into AI-assisted, parametric, and user-configurable prosthetic design platforms that can systematically address personalization while incorporating validated user involvement methodologies throughout the design lifecycle.

References

A U Iwuoha, and C. Iheme. "Design and 3d-Printing of a Body-Powered Prosthetic Limb for Amputee," 2020.

A. Hussaini, P. Kyberd, B. Mulindwa, R. Ssekitoleko, W. Keeble, L. Kenney, and David Howard. "3D Printing in LMICs: Functional Design for Upper Limb Prosthetics in Uganda." *Prosthesis*, 2023.

A. Roda-Sales, and Immaculada Llop-Harillo. "Identifying Users' Needs to Design and Manufacture 3D-Printed Upper Limb Sockets: A Survey-Based Study." *Applied Sciences*, 2024.

Albert Manero, John Sparkman, Matt Dombrowski, Peter Smith, Pavan Senthil, Spencer Smith, Viviana Rivera, and Albert Chi. "Evolving 3D-Printing Strategies for Structural and Cosmetic Components in Upper Limb Prosthesis." *Prosthesis*, 2023.

Alejandro M. Saldarriaga, Enzo Romero, Victoria E. Abarca, and Dante A. Elías. "A Parametric Design Approach for Affordable Customized 3D Socket for Transradial Upper Limb Prostheses." *International Conference on Control, Decision and Information Technologies*, 2024.

Amanda Coelho Figliolia, F. O. Medola, F. Sandnes, A. C. T. Rodrigues, and L. C. Paschoarelli. "Avoiding Product Abandonment Through User Centered Design: A Case Study Involving the Development of a 3D Printed Customized Upper Limb Prosthesis." *International Conference on Applied Human Factors and Ergonomics*, 2019.

Andrea Demofonti, G. Carpino, N. Tagliamonte, Giulia Baldini, Laura Bramato, and L. Zollo. "Design of a Modular and Compliant Wrist Module for Upper Limb Prosthetics." *The Anatomical Record*, 2022.

Bryce Dyer, Richard Glithro, and Abigail Batley. "The Design of an Upper Arm Prosthesis Utilising 3D Printing Conceived for the 2020 Tokyo Paralympic Games: A Technical Note." *Journal of Rehabilitation and Assistive Technologies Engineering*, 2022.

F. Pinheiro, and Francisco. "Development of a Functional Upper Limb Prosthesis," 2018.

Guisheng Xu, Liang Gao, K. Tao, Shengxiang Wan, Yuning Lin, Ao Xiong, Bin Kang, and H. Zeng. "Three-Dimensional-Printed Upper Limb Prosthesis for a Child with Traumatic Amputation of Right Wrist." *Medicine*, 2017.

Jennifer Mankoff, Saiph Savage, S. Eckert, C. Ngo, and G. Fiedler. "USER EXPERIENCES WITH TRADITIONAL AND 3D-PRINTED UPPER EXTREMITY PROSTHESES, DEVELOPMENT OF A COMPREHENSIVE SURVEY INSTRUMENT." *Canadian Prosthetics & Orthotics Journal*, 2018.

Juan Sebastian Cuellar, G. Smit, A. A. Zadpoor, and P. Breedveld. "Ten Guidelines for the Design of Non-Assembly Mechanisms: The Case of 3D-Printed Prosthetic Hands." *Proceedings of the Institution of Mechanical Engineers. Part H, Journal of Engineering in Medicine*, 2018.

Julio Fajardo, Victor Ferman, Ali Lemus, and E. Rohmer. "An Affordable Open-Source Multifunctional Upper-Limb Prosthesis with Intrinsic Actuation." *Workshop on Advanced Robotics and Its Social Impacts*, 2017.

Julio Fajardo, Victor Ferman, Diego Cardona, Guillermo Maldonado, Ali Lemus, and E. Rohmer. "Galileo Hand: An Anthropomorphic and Affordable Upper-Limb Prosthesis." *IEEE Access*, 2020.

Jung Wook Park, B. Greenspan, Taylor Tabb, Eric M Gallo, and A. Danielescu. "3D Printed Energy Return Elements for Upper Limb Sports Prosthetics." *Prosthesis*, 2023.

K. Wendo, Séverine Guisset, K. Sawadogo, Olivier Barbier, Arnaud Meunier, Axele Felloneau, Thierry Oquidam, et al. "Design and Validation of a Web-Based Exploratory Survey Investigating Qualified Professionals and Volunteers Using 3D Printing for Upper Limb Prosthetics: A Methodological Paper." *Prosthesis*, 2025.

Kai Xu, and Shengfeng Qin. "An Interdisciplinary Approach and Advanced Techniques for Enhanced 3D-Printed Upper Limb Prosthetic Socket Design: A Literature Review." *Actuators*, 2023.

M. Hofmann, Jeffrey Harris, Scott E. Hudson, and Jennifer Mankoff. "Helping Hands: Requirements for a Prototyping Methodology for Upper-Limb Prosthetics Users." *International Conference on Human Factors in Computing Systems*, 2016.

Maxwell Salazar, Ricardo Rosero, Oscar Zambrano, and Paola Portero. "Impact of 3D Printing Technology for the Construction of a Prototype of Low-Cost Robotic Arm Prostheses." *Advances in Mechanical Engineering*, 2024.

Muhammad Firas Zulkifli, N. M. Hashim, M. H. Ramlee, Y. Whulanza, and Abdul Halim Abdullah. "Development of Customized Passive Arm Prosthetics by Integrating 3D Printing and Scanning Technology." *Int. J. Online Biomed. Eng.*, 2023.

Pérez Dickson, and G. Sofía. "Protcom, Complementos Para Tu Prótesis. Diseño de Accesorios Para Prótesis de "FABRILAB", de Miembro Superior Por Debajo de Codo," 2019.

Pierre Moreau, Samar Ismael, Hatim Masadeh, Esraa Al Katib, Laetitia Viaud, Clara Nordon, and Safa T. Herfat. "3D Technology and Telemedicine in Humanitarian Settings." *The Lancet Digital Health*, 2020.

Renato Mio, Midori Sanchez, Q. Valverde, José Lara, and F. Rumiche. "Mechanical Testing Methods for Body-Powered Upper-Limb Prostheses: A Case Study." *Advances in Science, Technology and Engineering Systems*, 2019.

Ricardo Rodrigues, Daniel Miranda, Vítor Carvalho, and Demétrio Matos. "Design and Development of an EMG Upper Limb Controlled Prosthesis: A Preliminary Approach." *Actuators*, 2025.

Saeed Bahrami Moqadam, Ahmad Saleh Asheghabadi, and Jing Xu. "A Novel Approach to Material Detection for Blind Amputee Users With Hand Prosthetic." *IEEE Sensors Journal*, 2025.

Wilson Sutanto Tan, Christian Harito, Grasheli Kusuma Andhini, Marcel Martawidjaja, Nico Chainando, Muhammad Syafi'i, Ketut B. Putra, and Muhammad Syafrudin. "Color Modification of Silicone-Based Prosthetic Finger by 3D-Printed Mold." *Prosthesis*, 2024.